

# Compute Bipartite Network Metrics from Community Data

## Description

Extracts the binary community matrix from a 'data\_to\_fit' list, binarizes it (any value > 0 becomes 1), and computes whole-network metrics via 'bipartite::networklevel()'. Returns a tidy tibble with one row per metric.

## Usage

```
compute_network_metrics(  
  data_to_fit = NULL,  
  vec_indices = c("connectance", "nestedness", "modularity")  
)
```

## Arguments

**data\_to\_fit** A named list as returned by 'build\_jsdm\_fit\_input()'. Must contain the element 'data\_community\_to\_fit', a numeric matrix with samples as rows and taxa as columns.

**vec\_indices** Character vector of network-level indices to compute, passed directly to the 'index' argument of 'bipartite::networklevel()'. Default: 'c("connectance", "nestedness", "modularity")'.

## Details

The function treats the community matrix as a lower-level (samples)  $C \times$  upper-level (taxa) bipartite network. Only the binary incidence (presence/absence) is used: all values > 0 are set to 1 before passing to 'bipartite::networklevel()'.

If the binarized matrix has no positive entries (i.e. no species observed in any sample), an error is raised.

## Value

A 'tibble' with two columns:

**'metric'**

Character. Name of the network metric.

**'value'**

Numeric. Computed value of the metric.

## See Also

[[build\\_jsdm\\_fit\\_input\(\)](#)], [[compute\\_community\\_presence\\_absence\(\)](#)]